

Ubiquitous Sensing, Computing & Communication Laboratory

Quality Laboratory Manual

Experiment No. 05

**To Understand interfacing of Pi Camera Module Interface with
Raspberry Pi using Python**

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Title of Experiment: To Understand interfacing of Pi Camera Module Interface with Raspberry Pi using Python

Aim of Experiment: To interface Camera Module to raspberry pi and take pictures and perform video recording.

System Requirements – Raspberry pi 4 Board Win 10 and above OS, 4GB RAM, 2.33 GHz Processor.

Hardware:

- Raspberry Pi 4 Model B
- MicroSD card (with Raspberry Pi OS installed)
- Pi Camera Module (v1.3)
- Breadboard
- Jumper wires
- Power supply (5V, 3A USB-C adapter)

Software:

- Raspberry Pi OS
- Python 3
- RPi. GPIO library

Experiment Objectives:

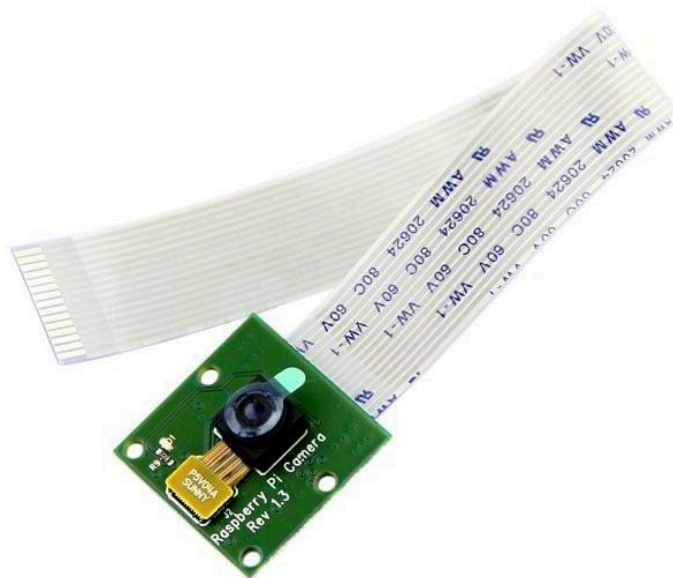
Objectives:

1. **To understand the interfacing and configuration of a camera module with Raspberry Pi** – Learn how to connect and enable the camera module using Raspberry Pi's CSI (Camera Serial Interface) port and configure it through the Raspberry Pi OS.
2. **To capture images and record videos using Raspberry Pi and Python programming** – Utilize Python scripts and command-line tools (e.g., `libcamera` or `picamera` library) to take pictures and record videos programmatically.
3. **To analyze and process the captured media for further applications** – Explore image processing techniques using OpenCV or other libraries to enhance, detect objects, or perform real-time video processing.

Experiment Outcomes:

- **Successful interfacing of the camera module with Raspberry Pi** – The camera module will be properly connected, configured, and recognized by the Raspberry Pi system.
- **Ability to capture images and record videos** – The Raspberry Pi will be able to take pictures and record videos using command-line tools or Python scripts.
- **Understanding of image processing and video handling** – The captured images and videos can be processed using Python libraries like OpenCV, enabling applications such as object detection, motion tracking, or real-time video streaming.

1.Theory:



Pi Camera Module (v1.3)

The pi Camera module is a camera that can be used to take pictures and high definition video.

Raspberry Pi Board has CSI (Camera Serial Interface) interface to which we can attach the PiCamera module directly.

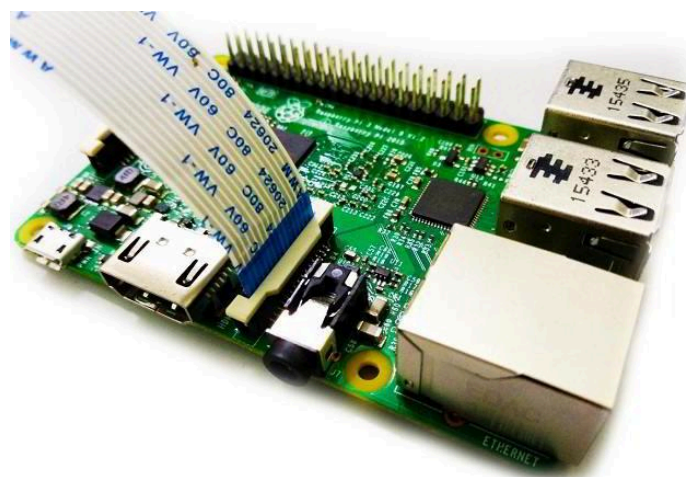
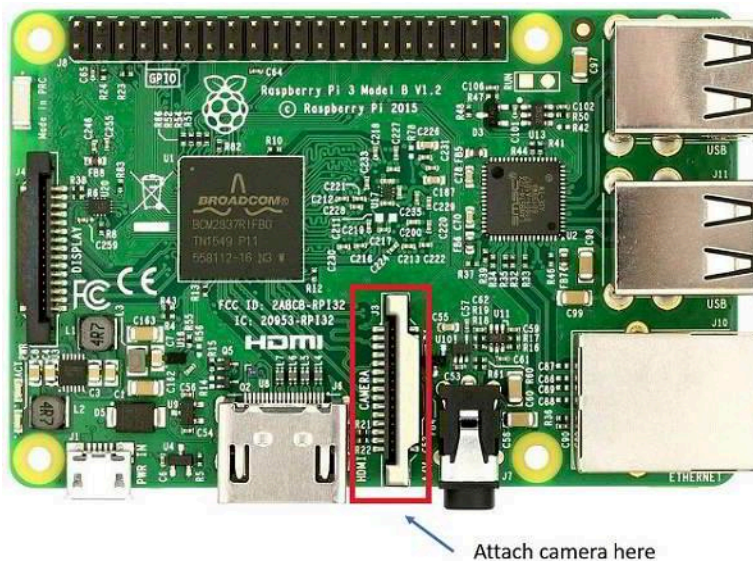
This Pi Camera module can attach to the Raspberry Pi's CSI port using a 15-pin ribbon cable.

Features of Pi Camera

Here, we have used Pi camera v1.3. Its features are listed below,

- Resolution – 5 MP
- HD Video recording – 1080p @30fps, 720p @60fps, 960p @45fps and so on.
- It Can capture wide, still (motionless) images of a resolution 2592x1944 pixels
- CSI Interface enabled.

How to attach Pi Camera to Raspberry Pi?



For enabling the camera in Raspberry Pi, open the raspberry pi configuration using the following command,

sudo raspi-config

then select Interfacing options in which select the camera option to enable its functionality.

reboot Raspberry Pi. Now we can access the camera on Raspberry Pi.

2. Example

Capture images and save it to the specified directory.

We can capture images using Python. Here, we will write a Python program to capture images using Pi Camera on Raspberry Pi.

Here, we have used **picamera** package(library) which provides different classes for Raspberry Pi. Out of which we are mainly interested in **PiCamera** class which is for camera module.

Pi Camera Python Program for Image Capture

```
import picamera
from time import sleep

#create object for PiCamera class
camera = picamera.PiCamera()
#set resolution
camera.resolution = (1024, 768)
camera.brightness = 60
camera.start_preview()
#add text on image
camera.annotate_text = 'Hi Pi User'
sleep(5)
#store image
camera.capture('image1.jpeg')
camera.stop_preview()
```

3.Conclusion:-

The **Raspberry Pi Camera Module** provides an efficient and versatile solution for capturing images and videos in embedded systems and IoT applications. By interfacing the camera via the **CSI port** and configuring it using **libcamera** or **picamera2**, users can seamlessly capture photos, record videos, and even stream live footage.

Expected Oral Questions:

1. What are the key differences between the Raspberry Pi Camera Module and a USB camera in terms of connectivity and performance?
2. How do you enable and test the Raspberry Pi Camera Module after connecting it to the CSI port?
3. What are the advantages of using `libcamera` over `raspistill` and `raspivid` for image and video capturing?
4. Can you explain how to capture an image using a Python script with the `picamera2` library?

FAQs in Interview:

1. How does the Camera Serial Interface (CSI) used by the Raspberry Pi Camera Module differ from USB in terms of data transfer and performance?
2. What steps would you take to troubleshoot a Raspberry Pi Camera Module that is not being detected by the system?
3. How can the Raspberry Pi Camera Module be used for real-time computer vision applications, and what libraries would you recommend?
4. Explain the process of setting up live streaming from a Raspberry Pi Camera Module. What software tools would you use?